



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WS-002**

Date **4-Jul-13**

Revision No. **1**

Supporting PQR No.(s) **RNC-PQR-002**

Date **23-Feb-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding (GTAW)**

(Automatic, **Manual**, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **GROOVE & FILLET**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1 or 2** to P-No. **1** Group No. **1 or 2**

Material Specification Type and Grade **A106 Gr.B, A105 Gr.B, A 234 WPB (or as per P-1)** TO **A106 Gr.B, A105 Gr.B, A 234 WPB (or as per P-1)**

Chem. Analysis and Mech. Prop. **-**

Thickness Range :

Base Metal : Groove **1.5 mm - 17.12 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other **N/A**

FILLER METALS (QW-404)

Spec. No. (SFA) **5.18**

AWS No. (Class) **ER 70S-6**

F-No. **6**

A-No. **1**

Size of Filler Metals **Ø 2.0 & 2.4 mm**

Deposited Weld Metals **8.56 mm.**

Thickness Range :

Groove **17.12 mm Max.**

Fillet **All**

Electrode-Flux (Class) **-**

Flux Trade Name **-**

Consumable Insert **-**

Other **Kobelco TGS-51T or Equivalents**

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**

Time Range **N/A**

Notes **N/A**

PREHEAT (QW-406)

Preheat Temp. Min.

Interpass Temp. Max. **30°C (Ambient)**

Preheat Maintenance **N/A**

Notes **N/A**

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	10-20 Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WPS No. RNC-WS-002 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity See Below
 Amps (Range) See Below Volts (Range) See Below
 Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead STRING OR WEAWE
 Orifice or Gas Cup Size NO. 4-9
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) See Below
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-6	2.4	DC EN	90-120	9-12	4-11	
2nd	GTAW	ER 70S-6	2.4	DC EN	100 -130	9-14	4-11	
Other(s)	GTAW	ER 70S-6	2.4	DC EN	100 -140	10-14	4-11	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of ASME IX, 2011 latest Revision.

Prepared by :

Mr Kittipong M.
Name

Signed

July 4, 2013
Date

Reviewed by :

Phuchin S.
Name

Signed

4/7/2013
Date

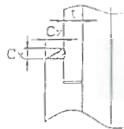
Approved by :

phuchin S.
Name

Signed

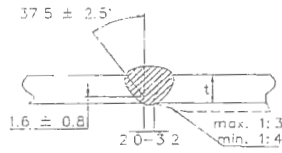
4/7/2013.
Date

ATTACHED FIGURE JOINT SKETCH



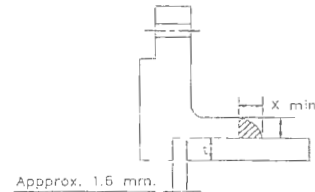
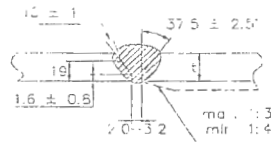
C_x (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm

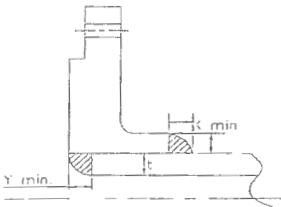
JOINT No. B1 & B2



Approx. 1.6 mm.

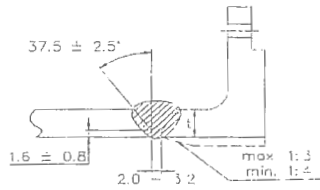
X min = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



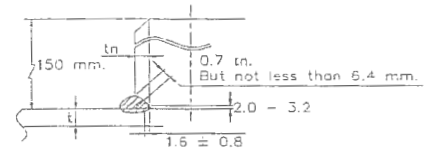
X min. = The lesser of $1.4t$ or thickness of the hub.
 Y min. = The lesser of t or 6.4 mm

JOINT No. A3



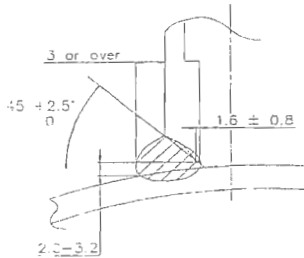
Misalignment (max.) = 1.5 mm

JOINT No. B3

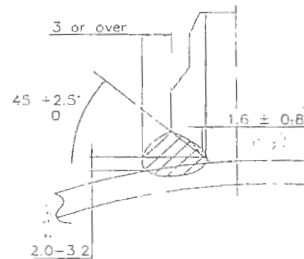


Misalignment (max.) = The lesser of 3.2 mm or $0.5t_n$.

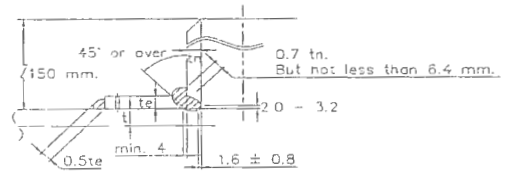
JOINT No. B4



JOINT No. C1

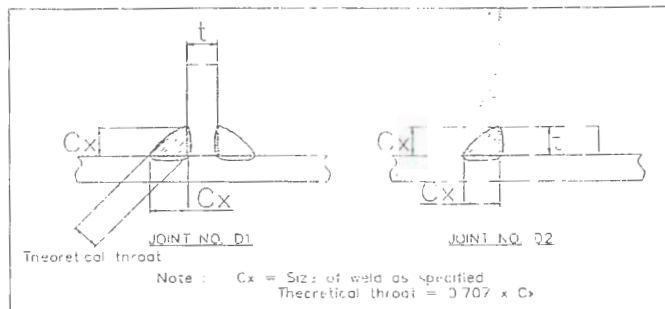


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm or $0.5t_n$

JOINT No. B5



Procedure Qualification Record (PQR)

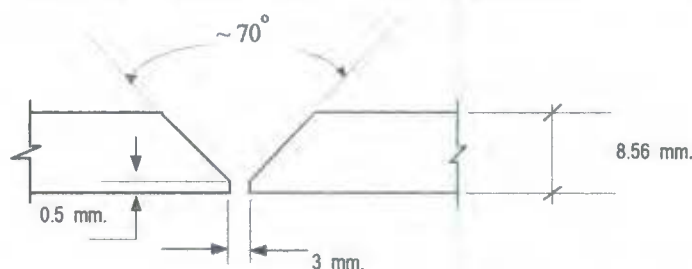
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-002

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-002	Date :	23 February 2013
WPS Number :	RNC-WS-002		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A 106	To	A 106	Temperature	N/A		
Type or Grade	Gr.B	To	Gr.B	Hold Time	N/A		
P. No.	1	To	1	Others			
Thickness of Test Coupon	8.56 mm.			GAS (QW-408)			
Diameter of Test Coupon	Ø 4"						
Others				Percent Composition			
					Gas (es)	Mixture	Flow Rate
				Shielding	Argon	99.99%	11 L/Min.
				Trailing	-	-	-
				Backing	-	-	-
FILLER METALS (QW-404)				ELECTRICAL CHARACTERISTICS (QW-409)			
SFA Specification	A 5.18			Current	DC		
AWS Classification	ER70S-G			Polarity	EN		
Filler Metal F-No.	6			Amperage	105 - 119	Voltage	10 - 10
Weld Metal Analysis A-No.	1			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Size of Filler Metal	Ø 2.4 mm.			Others	N/A		
Others	Kobelco TGS-50			TECHNIQUE (QW-410)			
Weld Metal Thickness	-						
POSITION (QW-405)				Travel Speed	4.97 - 9.28 cm/min.		
Position of Groove	6G			String or Weave Bead	Both		
Progression of Welding (Uphill or Down)	Up Hill			Oscillation	N/A		
Others	N/A			Multiple or single Pass	Multiple Pass		
PREHEAT (QW-406)				Multiple or Single Electrodes	Single Electrode		
Preheat Temperature	33° C			Others	N/A		
Interpass Temperature (max.)	179° C						
Others	N/A						

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number: RNC-PQR-002

Tensile Test Report No.TS-13-02-083

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
544/13 TR1	19.13	8.14	155.72	85.10	546.51	OUT OF WELD
544/13 TR2	19.11	8.28	158.230	85.80	542.22	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-032

Type and Figure Number	Type of Bend	Indication	Results
544/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
544/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
544/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
544/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-006

Hardness Testing Report No. HV-13-02-014

Macrostructure Report No. MA-13-02-048

Impact Testing Report No.IM-13-02-091

Welders Name Mr.Prathom M. Welder Number W-009
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by :

Certified by :

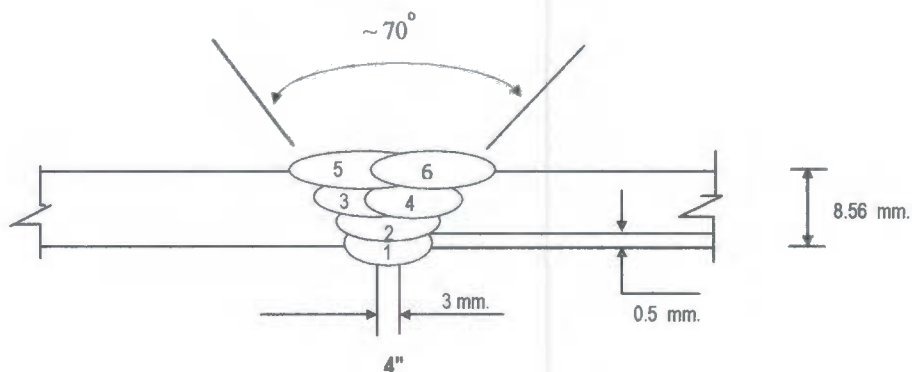
Date :



: 23 February 2013

	WELDING PROCEDURE	PQR No. : RNC-PQR-002
		WPS No. : RNC-WS-002
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for PQR-RNC-002 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER70S-G	2.4	105	10	DCEN	-	-	4.97	12.68	11	-
2	GTAW	ER70S-G	2.4	108	10	DCEN	-	-	9.28	6.99	11	-
3	GTAW	ER70S-G	2.4	106	10	DCEN	-	-	7.62	8.35	11	-
4	GTAW	ER70S-G	2.4	119	10	DCEN	-	-	7.36	9.71	11	-
5	GTAW	ER70S-G	2.4	118	10	DCEN	-	-	6.24	11.35	11	-
6	GTAW	ER70S-G	2.4	116	10	DCEN	-	-	6.10	11.41	11	-

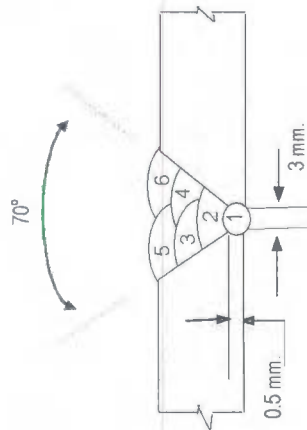
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

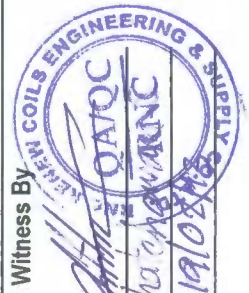
Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-018	
Customer Order No. : N/A		WPS No. : RNC-WPS-002		Date : 13-Feb-13			
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material Spec	Size	Electrode Specification
W-009	MR.PRATHOM M.	GTAW	6G	ASME IX	A106 Gr.B	ø 4" Thk. 8.56 mm.	A5.18
				Results	Visual	RT	
					ACCEPTED		ACCEPTED

Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backin g
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	
1	GTAW	ER70S-G	2.4	DCEN	105	10	4.97	33	12.68	Argon	11	-
2	GTAW	ER70S-G	2.4	DCEN	108	10	9.28	91	6.99	Argon	11	-
3	GTAW	ER70S-G	2.4	DCEN	106	10	7.62	103	8.35	Argon	11	-
4	GTAW	ER70S-G	2.4	DCEN	119	10	7.36	179	9.71	Argon	11	-
5	GTAW	ER70S-G	2.4	DCEN	118	10	6.24	150	11.35	Argon	11	-
6	GTAW	ER70S-G	2.4	DCEN	116	10	6.10	153	11.41	Argon	11	-

Joint Details



PAE Inspection By



Approved By

SIGNATURE :
NAME : Mr. Wichit K.
DATE : 14-Feb-2013

SIGNATURE :
NAME : Mr. Chaiyaporn RNC
DATE : 19/02/13

SIGNATURE :
NAME :
DATE :

Phuchin Seebomchai



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakk ong bangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

29779

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax 0 2815 6415, E-mail : qa@lpnplm.co.th

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-083

Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 4" (8.56 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment / Serial No.	INSTIRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	544/13 TR1	N/A	8.14	19.13	155.72	N/A	N/A	N/A	85.10	546.51	N/A	OUT OF WELD
2	544/13 TR2	N/A	8.28	19.11	158.23	N/A	N/A	N/A	85.80	542.22	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009



Reported by : N. Keerati Sa-ngoen

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : N. Pavaret Preedawiphat

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

29785

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-032

Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 4" (8.56 mm Thk.)	Diameter of Meudrel	32 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	544/13 BF1	Transverse face bend	8.14x19.14x160.73 mm	No Open Discontinuity	N/A	-
2	544/13 BF2	Transverse face bend	8.11x19.15x160.77 mm	No Open Discontinuity	N/A	-
3	544/13 BR1	Transverse root bend	8.14x19.17x160.46 mm	No Open Discontinuity	N/A	-
4	544/13 BR2	Transverse root bend	8.68x19.14x160.55 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009



Reported by : N. Sa-ngoen

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : N. Pavaret

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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LPN PLATE MILL



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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-02-091

Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a ^{cl}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	Ø 4" (8.56 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	19 February 2013	Equipment 's Capacity	542 Joules
Tested Date	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	544/13-1	-46°C	6.713	10.006	54.88	14.68	26.33	Weld Metal	-
2	544/13-2	-46°C	6.713	10.004	54.87	51.75			
3	544/13-3	-46°C	6.713	10.008	54.84	12.57			
4	544/13-1	-46°C	6.713	10.007	54.91	80.46	92.08	Base Metal	-
5	544/13-2	-46°C	6.714	10.008	54.86	100.51			
6	544/13-3	-46°C	6.713	10.010	54.90	95.27			
7	544/13-1	-46°C	6.713	10.008	54.95	175.12	167.41	HAZ	-
8	544/13-2	-46°C	6.714	10.009	54.98	179.75			
9	544/13-3	-46°C	6.714	10.008	54.96	147.36			

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009



Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date 22 / 02 / 13

Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date 22 / 02 / 13

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

13695

Macrostructure Analysis Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

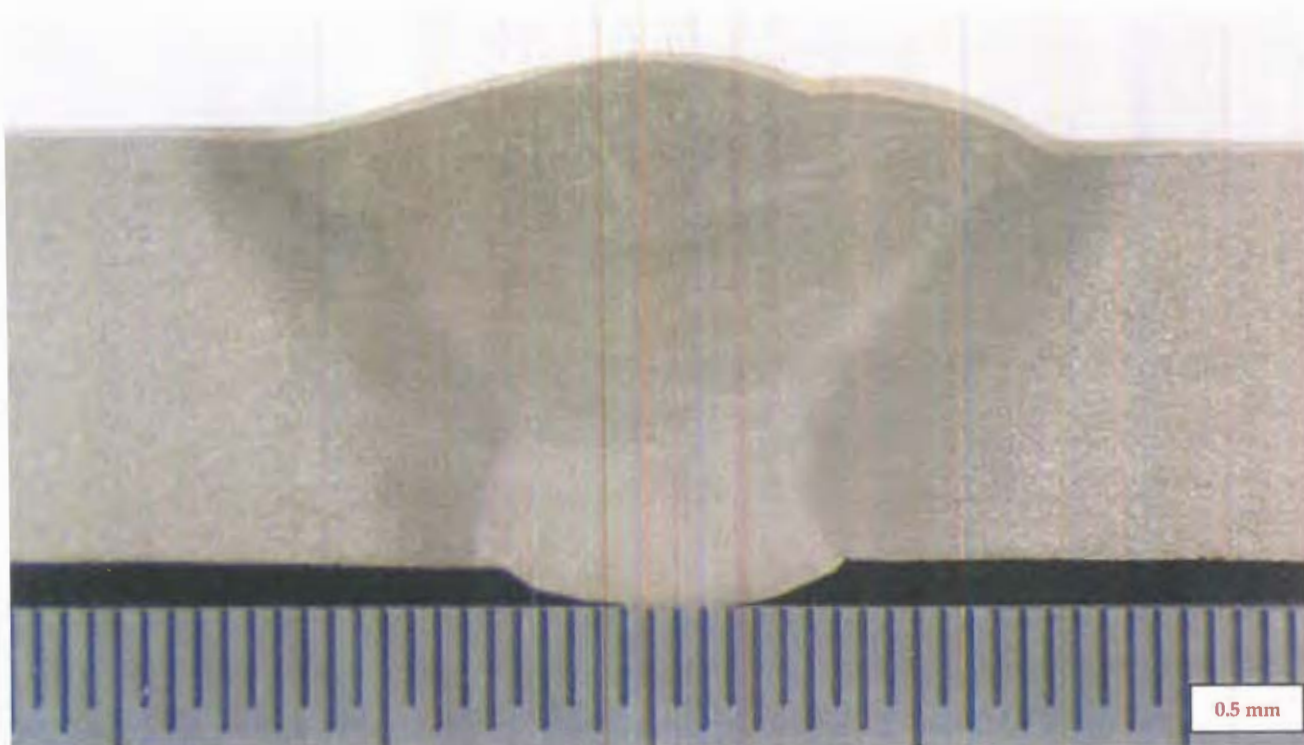
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

Report No. MA-13-02-048

Revision No.0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Test No.	544/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A106 Gr.B	Method of Preparation	ASME Secction IX : 2010 edition
Dimension	4" (8.56 mm Thk.)	Test Method	ASME Secction IX : 2010 edition
Welding Process/Position	GTAW/6G	Standard of Test	ASME Secction IX : 2010 edition
Type of Joint	Butt Jiont	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	Mr. Prathom M. / W-009	Time of Etching	5 minutes
Date Received	18 February 2013	Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by : 
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13



Approved by : 
(Mr.Pavaret Freedawiphat)
Deputy Managing Director
Date : 27/04/13

Notes

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LPN PLATE MILL



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

13231

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Report No. HV-13-02-014

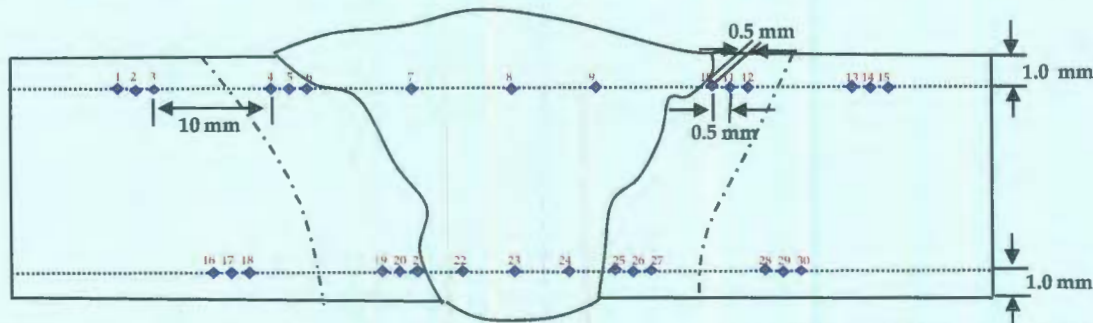
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Test No.	544/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Secction IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A106 Gr.B	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	4" (8.56 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Jiont	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0 °C

Test Location

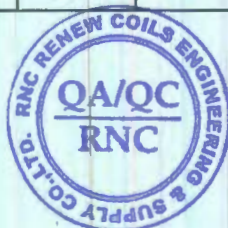
Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	147.5	16	BASE	154.8	*****		
2	BASE	148.3	17	BASE	152.5			
3	BASE	148.5	18	BASE	151.8			
4	HAZ	166.7	19	HAZ	162.2			
5	HAZ	176.4	20	HAZ	165.6			
6	HAZ	189.2	21	HAZ	169.2			
7	WELD	196.6	22	WELD	155.4			
8	WELD	191.2	23	WELD	154.6			
9	WELD	192.2	24	WELD	153.9			
10	HAZ	188.4	25	HAZ	164.6			
11	HAZ	177.4	26	HAZ	166.9			
12	HAZ	175.0	27	HAZ	162.0			
13	BASE	151.8	28	BASE	152.4			
14	BASE	152.4	29	BASE	152.5			
15	BASE	152.6	30	BASE	149.8			

Additional details : -

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22 / 02 / 13



Approved by :
(Mr.Pavaret Freedawiphat)
Deputy Managing Director
Date : 22 / 02 / 13

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